

Neil Malan

n.malan@unsw.edu.au | 0477143286

EDUCATION

UNIVERSITY OF CAPE TOWN
BSc (HONS, 1ST CLASS), MARINE
BIOLOGY AND OCEAN AND ATMOS.
SCIENCE

Dec 2011 | Cape Town, South Africa

**MSc, OCEAN AND CLIMATE
DYNAMICS**

Jul 2013 | Cape Town, South Africa

PHD, PHYSICAL OCEANOGRAPHY

Jul 2018 | Cape Town, South Africa

LINKS

Website:// nmalan.github.io

Github:// nmalan

LinkedIn:// neil-malan

RESEARCH STUDENTS

COMPLETED

Fernando Sobral. **PhD, UNSW 2026**

Cross-shelf heat transport in the East
Australian Current

Clea Welch. **MSc, UCT 2026**

Subsurface marine heatwaves in the
Mozambique Channel

Yotam Fadida. **MSc, UCT 2019**

Trends in the Agulhas Return Current
(published in DSR)

Caitlin Sole. **Hons, UCT 2016**

Impact of relative and absolute winds on
eddy variability.

CURRENT

Emma Hu - summer research intern
(UNSW) - 1D biogeochemical modelling of
the impacts of MHW vertical structures.

Erika Lehmann - Honours (UNSW) - A
Novel Lagrangian-Eulerian Framework To
Examine the Potential Migration of
Various Species Polewards Down the East
Australia Current

Rosa Fenhalls - Taught MSc thesis
(UNSW) - A machine learning approach to
predicting salinity in the Hawkesbury
estuary.

Siyang Liu - Taught MSc thesis (UNSW) -
Modelling sea urchin *Centrostephanus
rodgersii* larval dispersal from mainland
Australia to Tasmania under EAC
intensification.

RESEARCH PROFILE

I am an oceanographer with a background in boundary currents, eddies, cross-shelf exchange and mixing – and the observational and modelling methods used to quantify them. Increasingly I am drawn to the ecological side of these systems, where physical structure sets the conditions for productivity, habitat and dispersal. I work especially on marine extremes, including marine heatwaves and flood-driven salinity events, and on the methods and conceptual framing that link them to biological outcomes. Much of my contribution lies in connecting scales and disciplines – relating basin-scale variability to local coastal impact, and working across the boundaries between oceanography, ecology, fisheries science and coastal hazards.

EXPERIENCE

**UNIVERSITY OF NEW SOUTH WALES | SENIOR RESEARCH
ASSOCIATE, CLIMATE CHANGE RESEARCH CENTRE**

July 2023 - Present | Sydney, Australia

- Academic level B position, working with Prof. Alex Sen Gupta
- Research focus on subsurface marine heatwaves.

**UNIVERSITY OF NEW SOUTH WALES | POSTDOCTORAL RESEARCH
FELLOW, COASTAL AND REGIONAL OCEANOGRAPHY LABORATORY**

Nov 2018 - June 2023 | Sydney, Australia

- Academic level A postdoctoral position in the School of Biological, Earth and Environmental Sciences, working with Professor Moninya Roughan.
- Research focus on cross-shelf Western Boundary Current processes and temperature trends
- Contribute to management of IMOS observation programme as deputy-leader, New South Wales Moorings.

**NANSEN-TUTU CENTRE FOR MARINE ENVIRONMENTAL
RESEARCH. | POSTDOCTORAL RESEARCH FELLOW**

February 2018 – Sep 2018 | Cape Town, South Africa

- Assist in development of base HYCOM model for operational data assimilation.
- Supervise post-graduate research student.
- Organise international workshop on modelling the Agulhas Current System.

**SOUTH AFRICAN ENVIRONMENTAL OBSERVATION NETWORK |
DOCTORAL RESEARCH FELLOW - PROFESSIONAL DEVELOPMENT PROGRAM**
November 2013 – Sep 2017 | Cape Town, South Africa

- Undertake PhD research.
- Contribute marine expertise to national science meetings.
- Participate in educational programmes for underprivileged schools.

TEACHING

UNIVERSITY OF NEW SOUTH WALES | COURSE CO-CONVENOR

2024 - present | Sydney, Australia

- MSCI3001/MSCI5004 Physical Oceanography (with Prof. Alex Sen Gupta) – curriculum design, course management, lectures, tutorials and assessment development.
- 2025 overall student satisfaction 5.63/6, well above School and Faculty averages.

RHODES UNIVERSITY | LECTURER, DEPT. OF ICHTHYOLOGY & FISHERIES SCIENCE

2012 - 2013 | Grahamstown, South Africa

- Taught and coordinated the Fisheries Oceanography module for 2nd- and 3rd-year students.

TEACHING ASSISTANT | UNIVERSITY OF CAPE TOWN & SYDNEY INSTITUTE OF MARINE SCIENCE

2014 - 2017; 2022

- Ocean Dynamics (2nd year) and Operational Oceanography (MSc), UCT.
- Topics in Australian Marine Science (MSc), SIMS.

AWARDS

- 2023 Invited Speaker, *Gordon Research Seminar on Coastal Ocean Dynamics*
- 2021 Best Postdoctoral Research Paper, *UNSW CMSI*
- 2018 Invited participant, Physical Oceanography Dissertation Symposium, *National Science Foundation*
- 2016 Best Student Presentation, Agulhas Current Research Symposium, *SAEON*
- 2015 SPACES travel fellowship (2-month funded research visit), *GEOMAR, Kiel*

ACADEMIC SERVICE AND OUTREACH

- **Member** 2023-present, Marine Heatwaves International Working Group.
- **Member of Australia's Marine National Facility scientific advisory committee** 2024-present, provide expert advice to the Marine National Facility (MNF) Steering Committee, primarily focused on assessing the research quality of proposed MNF voyages and other matters requiring scientific expertise.
- **Convenor** Oceania Marine Heatwave Seminars. Jan 2025 - present
- **Member of the scientific committee**, 2026 International Conference on Southern Hemisphere Meteorology and Oceanography.
- **Invited contributor**, three chapters (one as co-lead), AGU monograph on marine heatwaves and their impacts (in preparation).
- **Deputy Leader IMOS NSW moorings 2018-2023**, assist in management of mooring program, including contribution to national committees, meeting reporting requirements at both state and national levels, tracking impact and developing data analysis tutorials.
- **Session Convenor or co-convenor** for sessions on Australian Boundary Currents at the Australian Meteorological and Oceanographic Society Annual meetings in 2020 and 2022, as well as Subtropical Western Boundary Current - Shelf interactions at the Ocean Sciences Meeting in 2022.
- **Expert Reviewer** for over 40 articles in journals such as *Geophysical Research Letters*, *Nature Climate Change*, *Journal of Physical Oceanography*, *Global Biogeochemical Cycles* and *Geoscientific Model Development*.

PUBLICATIONS

I have published widely in top-tier oceanography and climate journals, with 10 first-author and 5 second-author papers. My work has been cited 535 times with an h-index of 13 (Google Scholar, 11/06/2026).

- [1] Neil Malan, Alex Sen Gupta, Amandine Schaeffer, Shujing Zhang, Martina A Doblin, Gabriela Semolini Pilo, Andrew E Kiss, Jason D Everett, Erik Behrens, Antonietta Capotondi, et al. "Lifting the lid on Marine Heatwaves". In: *Progress in Oceanography* (2025), p. 103539.
- [2] Kathryn E Smith, Alex Sen Gupta, Dillon Amaya, Jessica A Benthuisen, Michael T Burrows, Antonietta Capotondi, Karen Filbee-Dexter, Thomas L Frölicher, Alistair J Hobday, Neil J Holbrook, et al. "Baseline matters: Challenges and implications of different marine heatwave baselines". In: *Progress in Oceanography* 231 (2025), p. 103404.
- [3] Kathryn E. Smith, Alex Sen Gupta, Michael T. Burrows, Karen Filbee-Dexter, Alistair J. Hobday, Neil J. Holbrook, Neil Malan, Pippa J. Moore, Eric C. J. Oliver, Mads Thomsen, Thomas Wernberg, Zijie Zhao, and Dan A. Smale. "Ocean extremes as a stress test for marine ecosystems and society". In: *Nature Climate Change* 15.3 (2025), pp. 231–235.

- [4] Clea B Welch, Neil Malan, Daneeja Mawren, Tamaryn Morris, Janet Sprintall, and Juliet C Hermes. "Subsurface manifestation of marine heat waves in the southwestern Indian Ocean". In: *Ocean Science* 21.4 (2025), pp. 1695–1708.
- [5] Neil Malan, Moninya Roughan, Michael Hemming, and Tim Ingleton. "Quantifying coastal freshwater extremes during unprecedented rainfall using long timeseries multi-platform salinity observations". In: *Nature Communications* 15.1 (2024), p. 424.
- [6] Fernando Sobral, Moninya Roughan, Neil Malan, and Junde Li. "Mean-state and seasonal variability in temperature structure and heat transport in the East Australian Current System from a multi-decadal regional ocean model". In: *Journal of Geophysical Research: Oceans* 129.9 (2024), e2023JC020438.
- [7] Michael P Hemming, Moninya Roughan, Neil Malan, and Amandine Schaeffer. "Observed multi-decadal trends in subsurface temperature adjacent to the East Australian Current". In: *Ocean Science* 19.4 (2023), pp. 1145–1162.
- [8] Neil Malan, Moninya Roughan, Michael Hemming, and Amandine Schaeffer. "Mesoscale circulation controls chlorophyll concentrations in the East Australian Current separation zone". In: *Journal of Geophysical Research: Oceans* 128.4 (2023), e2022JC019361.
- [9] Neil Malan, Moninya Roughan, Geoffrey J Stanley, Ryan Holmes, and Junde Li. "Quantifying cross-shelf transport in the East Australian Current system: a budget-based approach". In: *Journal of Physical Oceanography* 52.10 (2022), pp. 2555–2572.
- [10] Lachlan R Phillips, Neil Malan, Moninya Roughan, Robert Harcourt, Ian Jonsen, Martin Cox, Andrew S Brierley, David Slip, Adam Wilkins, and Gemma Carroll. "Coastal seascape variability in the intensifying East Australian Current Southern Extension". In: *Frontiers in Marine Science* (2022).
- [11] Hayden T Schilling, Daniel E Hewitt, Neil Malan, Matthew D Taylor, and Daniel D Johnson. "Cross-jurisdictional larval supply essential for eastern Australian spanner crabs (*Ranina ranina*)". In: *Marine and Freshwater Research* 73.11 (2022), pp. 1352–1367.
- [12] Yotam Fadida, Neil Malan, Meghan F Cronin, and Juliet Hermes. "Trends in the Agulhas Return Current". In: *Deep Sea Research Part I: Oceanographic Research Papers* 175 (2021), p. 103573.
- [13] Neil Malan, Moninya Roughan, and Colette Kerry. "The rate of coastal temperature rise adjacent to a warming western boundary current is nonuniform with latitude". In: *Geophysical Research Letters* 48.3 (2021), e2020GL090751.
- [14] Neil Malan, Matthew Archer, Moninya Roughan, Paulina Cetina-Heredia, Michael Hemming, Carlos Rocha, Amandine Schaeffer, Iain Suthers, and Eduardo Queiroz. "Eddy-driven cross-shelf transport in the East Australian Current separation zone". In: *Journal of Geophysical Research: Oceans* 125.2 (2020), e2019JC015613.
- [15] Jethan Shaun d'Hotman, Neil Malan, Charine Collins, Marc de Vos, Rick Lumpkin, Tamaryn Morris, and Juliet Hermes. "The use of a jet reference frame to analyze drifter trajectories in the Agulhas Current". In: *Journal of Geophysical Research: Oceans* 124.6 (2019), pp. 4238–4247.
- [16] Neil Malan, Jonathan V Durgadoo, Arne Biastoch, Chris Reason, and Juliet Hermes. "Multidecadal wind variability drives temperature shifts on the Agulhas Bank". In: *Journal of Geophysical Research: Oceans* 124.5 (2019), pp. 3021–3035.
- [17] Ahmad Fehmi Dilmahamod, B Aguiar-González, Pierrick Penven, CJC Reason, WPM De Ruijter, N Malan, and JC Hermes. "SIDDIES corridor: a major east-west pathway of long-lived surface and subsurface eddies crossing the subtropical South Indian Ocean". In: *Journal of Geophysical Research: Oceans* 123.8 (2018), pp. 5406–5425.
- [18] N. Malan. "Are icebergs a realistic option for augmenting Cape Town's water supply?" In: *Water Wheel* 17.2 (2018), pp. 32–34.
- [19] Neil Malan, Björn Backeberg, Arne Biastoch, Jonathan V Durgadoo, Annette Samuelsen, Chris Reason, and Juliet Hermes. "Agulhas Current Meanders facilitate shelf-slope exchange on the Eastern Agulhas Bank". In: *Journal of Geophysical Research: Oceans* (2018).
- [20] N Malan, CJC Reason, and BR Loveday. "Variability in tropical cyclone heat potential over the Southwest Indian Ocean". In: *Journal of Geophysical Research: Oceans* 118.12 (2013), pp. 6734–6746.

PREPRINTS, IN-REVIEW AND SUBMITTED PAPERS

- Karen Filbee-Dexter, Lianna Gendall, Albert Pessarrodona, Michael T Burrows, Alex Sen Gupta, Alistair J Hobday, Neil J Holbrook, Neil Malan, Shinae Montie, Kira A Krumhansl, Pippa J Moore, Erik C J Oliver, Dan A Smale, Kathryn E Smith, Samuel A Starko, Mads S Thomsen, Mathew A Vanderklift, Zijie Zhao, and Thomas Wernberg. "Marine heatwaves drive widespread loss and slow recovery of coastal carbon cycling". In: *Communications Earth & Environment (submitted)* (2026). Preprint posted on Research Square, 28 May 2026. doi: 10.21203/rs.3.rs-9834051/v1.
- Ryan M Holmes and Neil Malan. "Accurately diagnosing mixed layer tracer budgets and marine heatwave drivers in ocean models". In: *Journal of Advances in Modeling Earth Systems (submitted)* (2026). Preprint on ESSOAr; submitted 10 June 2026.
- Neil Malan, Moninya Roughan, Michael Hemming, Neil Doszpot, Colin L Johnson, Samantha K Lynch, Robert A B Mason, and Tim Ingleton. "Observations of river plumes in southeastern Australia impacting the coastal ocean following extreme rainfall in 2022". In: *Marine and Freshwater Research (in review)* (2026).

- Kathryn E Smith, Julio Salcedo-Castro, Neil J Holbrook, Serge Andrefouet, Hans Burchard, Michael T Burrows, Bronwyn Cahill, Sophie Cravatte, Karen Filbee-Dexter, David Carrasco, Christopher D G Harley, Alistair J Hobday, Nathan G King, Romain Le Gendre, Neil Malan, Murilo Zanetti Marochi, Cristian Martinez-Villalobos, Piero L F Mazzini, Pippa J Moore, Antonio Olita, Eric Oliver, Andrea Piñones, Cécile Pujol, Alex Sen Gupta, Spencer Tassone, Mads S Thomsen, Ryan K Walter, Thomas Wernberg, and Dan A Smale. “Marine heatwaves in coastal systems”. In: *Nature Reviews Earth & Environment (in review)* (2026).

REFEREES

- Prof. Alex Sen Gupta (UNSW) a.sengupta@unsw.edu.au
- Prof. Iain Suthers (UNSW) i.suthers@unsw.edu.au
- Dr Alistair Hobday (CSIRO) alistair.hobday@csiro.au